# Pushing the Limits of Novel View Synthesis in Soccer Scenes Through Ensembles

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## Abstract

*We, the Hands-On Computer Vision team, propose a novel view synthesis pipeline for soccer scenes. We enhance the baseline along two axes: monocular depth priors combined with an improved appearance module to guide geometric reconstruction, and an ensemble that aggregates multiple renderings to reduce variance under extreme novel viewpoints. Additionally, we apply a final color-matching post-processing step. Our code will be publicly available at: https://github.com/cvail-research/soccernet-nvs-2026.*

## 1. Proposed Method

### 1.1. Overview

Our method builds upon the 3D Gaussian Splatting [1] framework, using the open-source *gsplat* [2] library from Nerfstudio as our backbone, and enhances it along two complementary axes. First, we incorporate monocular depth priors to guide the geometric reconstruction. Second, we use a model ensembling strategy that aggregates the predictions of several independently trained renderings, each contributing its respective strengths.

### 1.2. Architecture

#### 1.2.1. Appearance Compensation Module

We enhance the appearance module in *gsplat* [2] by incorporating feature normalization, regularization, and a residual learning formulation. Specifically, for a camera $c$ and a Gaussian $i$, let $\mathbf{e}_c \in \mathbb{R}^{d_e}$ denote the camera-specific embedding, $\mathbf{f}_i \in \mathbb{R}^{d_f}$ the per-Gaussian latent feature vector, and $\mathbf{SH}_\ell(\mathbf{d}_{c,i}) \in \mathbb{R}^K$ the spherical harmonics basis of degree $\ell$ evaluated at the viewing direction $\mathbf{d}_{c,i} \in \mathbb{R}^3$. We first apply stochastic regularization to the camera embedding, $\tilde{\mathbf{e}}_c = \mathrm{Dropout}(\mathbf{e}_c)$ and concatenate the regularized embedding with the remaining inputs to form the input:

$$\mathbf{h}_{c,i} = [\tilde{\mathbf{e}}_c \,;\, \mathbf{f}_i \,;\, \mathbf{SH}_\ell(\mathbf{d}_{c,i})] \in \mathbb{R}^D, \tag{1}$$

where $D = d_e + d_f + K$. Finally, we apply layer normalization to obtain the input passed to the appearance MLP, $\tilde{\mathbf{h}}_{c,i} = \mathrm{LN}(\mathbf{h}_{c,i})$. The appearance network is modeled as a multi-layer perceptron $g_\theta$:

$$\Delta\mathbf{c}_{c,i} = \tanh\left(g_\theta(\tilde{\mathbf{h}}_{c,i})\right) \odot \mathbf{s}, \tag{2}$$

where $\Delta\mathbf{c}_{c,i} \in \mathbb{R}^3$ is a residual color correction, $\mathbf{s} \in \mathbb{R}^3$ is a learnable per-channel scaling, and $\odot$ denotes element-wise multiplication. Let $\mathbf{C}_{c,i}^{\text{base}} \in \mathbb{R}^3$ denote the learnable per-Gaussian colors. The final color $\mathbf{C}_{c,i} \in [0,1]^3$ is obtained by adding the residual correction $\mathbf{C}_{c,i} = \mathbf{C}_{c,i}^{\text{base}} + \Delta\mathbf{c}_{c,i}$.

#### 1.2.2. Depth Prior Integration

Instead of relying solely on Structure-from-Motion (SfM) initialization from COLMAP [3], we employ relative monocular depth predictions obtained from Depth Anything V3 $\mathcal{D}(\cdot)$ (DA3-Large 0.36B parameters) [4]. These depth maps provide dense geometric priors to guide the reconstructions. For each training image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ we use $\mathcal{D}(\cdot)$ to produce dense depth maps $\mathbf{Y} = \mathcal{D}(\mathbf{I}) \in \mathbb{R}^{H\times W}$.

#### 1.2.3. Loss Function

Given the rendered depth map $\hat{\mathbf{Y}} \in \mathbb{R}^{H\times W}$ and ground truth $\mathbf{Y} \in \mathbb{R}^{H\times W}$, we first compute an optimal scale $s$ and shift $t$ such that $\hat{\mathbf{Y}}^* = s\hat{\mathbf{Y}} + t$ where $(s,t)$ are obtained by minimizing the squared error between $\mathbf{Y}$ and $\hat{\mathbf{Y}}^*$. The depth loss is then defined as:

$$\mathcal{L}_{\text{depth}} = \frac{1}{2}\left\|\mathbf{Y} - \hat{\mathbf{Y}}^*\right\|_2^2. \tag{3}$$

We also introduce a gradient-based regularization term on the residual $\mathbf{R} = \mathbf{Y} - \hat{\mathbf{Y}}^*$: The gradient loss is defined as:

$$\mathcal{L}_{\text{grad}} = \left(\|\nabla_x\mathbf{R}\|_1 + \|\nabla_y\mathbf{R}\|_1\right), \tag{4}$$

where $\nabla_x$ and $\nabla_y$ denote discrete spatial derivatives along the horizontal and vertical directions, respectively.

We adopt the standard photometric loss $\mathcal{L}_{\text{photo}}$ from 3D Gaussian Splatting[1]. The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{photo}} + \lambda_d\left(\mathcal{L}_{\text{depth}} + \lambda_g\mathcal{L}_{\text{grad}}\right), \tag{5}$$

where $\mathcal{L}_{\text{depth}}$ enforces alignment with the monocular depth, and $\mathcal{L}_{\text{grad}}$ promotes consistency of geometric structures.

### 1.3. Ensemble

We observe that different reconstruction models evaluated in extreme novel views exhibit complementary strengths: our 3DGS model faithfully reconstructs the largely static stadium background, whereas a model trained under the Triangle Splatting [5] formulation produces noticeably sharper renderings of the players and the ball. To leverage the best of both, we combine their predictions through a weighted ensemble in which the per-pixel weights are guided by the semantic content of each rendered view. To identify these regions, given a rendered novel view $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times 3}$ produced by a trained model, we run inference with a pre-trained segmentation network $f_\theta$ (SAM3 [6]) and obtain a binary mask

$$\mathcal{M} = f_\theta(\hat{\mathbf{X}}) \in \{0,1\}^{H\times W\times|\mathcal{R}|}, \tag{6}$$

that isolates the foreground classes $\mathcal{R} = \{\text{ball}, \text{player}_1, \text{player}_2, \ldots, \text{player}_{N_p}\}$ from the rest of the scene. We then collapse $\mathcal{M}$ across the class dimension into a single binary foreground indicator

$$\mathbf{M}_{h,w} = \max_{r\in\{1,\ldots,|\mathcal{R}|\}} \mathcal{M}_{h,w,r} \in \{0,1\}^{H\times W}, \tag{7}$$

beyond combining different model families, we further reduce the prediction error in image space by averaging multiple independently trained renderings. For $N$ renderings $\{\hat{\mathbf{X}}_n\}_{n=1}^N$ whose errors with respect to the ground-truth view $\mathbf{X}^*$ are zero-mean and independent with variance $\sigma^2$, the per-pixel error variance of their mean satisfies:

$$\text{Var}\left[\frac{1}{N}\sum_{n=1}^{N}\hat{\mathbf{X}}_n - \mathbf{X}^*\right] = \frac{\sigma^2}{N}, \tag{8}$$

This is especially relevant in the extreme novel-view regime, where each individual rendering exhibits its largest errors. Concretely, we train $N_g = 5$ 3DGS renderings $\{\hat{\mathbf{X}}_n^{(g)}\}_{n=1}^{N_g}$ together with a single Triangle Splatting rendering $\hat{\mathbf{X}}^{(t)}$, and we denote the 3DGS average by

$$\bar{\mathbf{X}}^{(g)} = \frac{1}{N_g}\sum_{n=1}^{N_g}\hat{\mathbf{X}}_n^{(g)}. \tag{9}$$

We then assign a per-pixel weight map to each rendering. The Triangle Splatting rendering is weighted only on foreground pixels with $w_t = 0.5$ we set $\mathbf{w}^{(t)} = w_t \cdot \mathbf{M}$, i.e. it receives a weight of $0.5$ where the mask is active and $0$ elsewhere, while the 3DGS average receives the complementary weight $\mathbf{w}^{(g)} = \mathbf{1} - \mathbf{w}^{(t)}$, with $\mathbf{1}$ denoting the all-ones $H \times W$ map. The final ensembled rendering is

$$\hat{\mathbf{X}}_{\text{ens}} = \mathbf{w}^{(t)} \odot \hat{\mathbf{X}}^{(t)} + \mathbf{w}^{(g)} \odot \bar{\mathbf{X}}^{(g)}, \tag{10}$$

| | gsplat | | Ours | |
|---|---|---|---|---|
| **Scene** | PSNR ↑ | SSIM ↑ | PSNR ↑ | SSIM ↑ |
| Scene 1 | **30.7053** | **0.8229** | 29.8623 | 0.8146 |
| Scene 2 | 28.2402 | 0.8043 | **30.1390** | **0.8414** |
| Scene 3 | 26.5945 | 0.7814 | **29.1802** | **0.8417** |
| Scene 4 | 28.5718 | 0.8130 | **28.6485** | **0.8246** |
| Scene 5 | 28.6487 | 0.7864 | **29.0613** | **0.8010** |
| **Average** | 28.5521 | 0.8016 | **29.3783** | **0.8247** |

Table 1. Comparison between gsplat baseline and our method in a synthetic validation set. Best results per scene are shown in bold.

## 2. Experimental Results

Since the official challenge split is held out, we build a synthetic validation set by holding out $20\%$ of the training images per scene. To mirror the viewpoint distribution of the challenge split, half of the held-out frames correspond to broadcast-style views and the other half to extreme close-up views.

**Implementation details.** All 3DGS models are trained for 80k iterations with AdamW on a single NVIDIA RTX PRO 6000 Blackwell GPU using the modified gsplat [2] framework, we set the loss weights to $\lambda_d = 10^{-3}$ and $\lambda_g = 0.5$, and keep all other hyperparameters at their default values. Each model is trained in 1 hour. The Triangle Splatting model is trained with the public official code [5].

Table 1 reports per-scene PSNR and SSIM on our validation split. Our method consistently outperforms the *gsplat* baseline, winning $4$ out of $5$ scenes on both metrics and raising the averages by $+0.83$ dB in PSNR ($28.55 \rightarrow 29.38$) and $+0.023$ in SSIM ($0.802 \rightarrow 0.825$). The largest gains concentrate on Scene 2 ($+1.90$ dB) and Scene 3 ($+2.59$ dB), where the baseline geometry is weakest, indicating that our depth-prior supervision and appearance-compensation module are most beneficial when SfM provides limited geometric constraints. The only exception is Scene 1, where the baseline retains a small advantage.

For the final challenge submission, we re-train each model on the full set of training views and apply a per-rendering color-matching post-processing step. Specifically, we transfer the color statistics of the training views onto every rendered image using the Multi-Variate Gaussian Distribution (MVGD) transfer [7], an analytical color-matching method that aligns the per-channel mean and covariance of the rendering with those of a reference image drawn from the training views. This step mitigates the residual color drift that 3DGS-based pipelines tend to exhibit under extreme novel viewpoints, where the appearance module extrapolates to camera embeddings far from the training distribution.

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